📘 Excel Intermediate Functions Documentation

Module: Logical, Text & Nested Functions in Excel

Using **Microsoft Excel**

🎯 Learning Objectives

By the end of this session, learners will be able to:

- Use logical functions for decision-making
- Manipulate text using text functions
- Build nested formulas
- Combine multiple functions for real-time business scenarios

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A. Logical Functions

Logical functions help Excel make decisions based on conditions.

1️⃣ IF Function

🔹 Purpose

Checks a condition and returns different results.

🔹 Syntax

```None
=IF(condition, value_if_true, value_if_false)
```

🔹 Example 1: Pass or Fail

Marks	Result
85	Pass
30	Fail

Formula:

```None
=IF(A2>=35,"Pass","Fail")
```

🔹 Explanation

Part	Meaning
A2>=35	Condition
"Pass"	If TRUE
"Fail"	If FALSE

🧪 Practice Activity

Create:

Salary	Status

25000	Eligible
12000	Not Eligible

Formula:

```None
=IF(A2>=20000,"Eligible","Not Eligible")
```

2️⃣ AND Function

🔹 Purpose

Returns TRUE only if ALL conditions are true.

🔹 Syntax

```None
=AND(condition1, condition2)
```

🔹 Example

Attendance	Marks	Result
80	70	Eligible

Formula:

```None
=AND(A2>=75,B2>=35)
```

🔹 Result

Conditions	Output
Both true	TRUE
One false	FALSE

3️⃣ OR Function

🔹 Purpose

Returns TRUE if ANY one condition is true.

🔹 Syntax

```
None
=OR(condition1, condition2)
```

🔹 Example

Sports	NCC	Selected
Yes	No	TRUE

Formula:

```
None
=OR(A2="Yes",B2="Yes")
```

4️⃣ NOT Function

🔹 Purpose

Reverses the result.

🔹 Syntax

```
None
=NOT(condition)
```

🔹 Example

```
None
=NOT(A2>100)
```

🔹 Result

Condition	Output
TRUE	FALSE
FALSE	TRUE

🔥 Real-Time Business Example

Employee Bonus Eligibility

Conditions:

- Salary > 30000
- Attendance > 90

Formula:

```None
=IF(AND(B2>30000,C2>90),"Bonus","No Bonus")
```

B. Text Functions

Text functions help clean and manipulate text data.

1️⃣ CONCAT / CONCATENATE

🔹 Purpose

Joins multiple text values.

🔹 Syntax

```None
=CONCAT(A2," ",B2)
```

🔹 Example

First Name	Last Name	Full Name
Tota	Jagadeesh	Tota Jagadeesh

Formula:

```None
=CONCAT(A2," ",B2)
```

2️⃣ LEFT Function

🔹 Purpose

Extracts characters from left side.

🔹 Syntax

```None
=LEFT(text, number_of_characters)
```

🔹 Example

```None
=LEFT("Excel",2)
```

Result:

```None
Ex
```

3️⃣ RIGHT Function

🔹 Purpose

Extracts characters from right side.

🔹 Example

```None
=RIGHT("Excel",2)
```

Result:

```None
el
```

4️⃣ MID Function

🔹 Purpose

Extracts text from middle.

🔹 Syntax

```None
=MID(text,start_num,num_chars)
```

🔹 Example

```None
=MID("Hospital",3,4)
```

Result:

```None
spit
```

5️⃣ LEN Function

🔹 Purpose

Counts total characters.

🔹 Example

```None
=LEN("Excel")
```

Result:

```None
5
```

6️⃣ TRIM Function

🔹 Purpose

Removes extra spaces.

🔹 Example

```None
=TRIM("   Ravi Kumar   ")
```

Result:

```None
Ravi Kumar
```

🔥 Important in Data Cleaning

Used heavily in:

- Employee names
- Imported datasets
- SQL exports

7️⃣ UPPER Function

Converts text to uppercase.

Example:

```None
=UPPER("excel")
```

Result:

```None
EXCEL
```

8️⃣ LOWER Function

Converts text to lowercase.

Example:

```None
=LOWER("EXCEL")
```

Result:

```None
excel
```

9️⃣ PROPER Function

Capitalizes first letter.

Example:

```None
=PROPER("tota jagadeesh")
```

Result:

```None
Tota Jagadeesh
```

🧪 Practice Activity

Raw Name	Clean Name
tota jagadeesh	Tota Jagadeesh

Use:

```None
=PROPER(TRIM(A2))
```

C. Nested Functions and Formula Building

🔹 What is a Nested Function?

A function inside another function.

Example

```None
=IF(A2>=35,"Pass","Fail")
```

Now combine with AND:

```None
=IF(AND(A2>=35,B2>=75),"Eligible","Not Eligible")
```

🔥 Real-Time Example

Employee Appraisal

Conditions:

- Salary > 30000
- Experience > 5 years

Formula:

```
None
=IF(AND(B2>30000,C2>5),"Promotion","No Promotion")
```

🔥 Combining Text Functions

Example:
Convert messy names into proper format.

Formula:

```
None
=PROPER(TRIM(A2))
```

🔥 Nested IF Example

Marks	Grade
90	A
70	B
40	C

Formula:

```
None
=IF(A2>=85,"A",IF(A2>=60,"B","C"))
```

🔍 Explanation

Condition	Grade

>=85	A
>=60	B
Else	C